

练习三十二 量子力学基础（二）

1. 0.0123nm

2. $\sqrt{6}\hbar, \pm 2\hbar, \pm \hbar, 0$

3. (3)

4. (4)

5. $hc/\lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{5.5 \times 10^{-7} \times 1.6 \times 10^{-19}} = 2.26(\text{ev})$

6. (1) $\int_0^a \psi^2(x)dx = 1 \quad A = \sqrt{\frac{2}{a}}$

(2) $\int_0^{a/4} \frac{2}{a} \sin^2 \frac{2\pi x}{a} dx = \frac{1}{4}$

(3) $E_2 = \frac{4h^2}{8ma^2} = \frac{h^2}{2ma^2}$